Practice Exam: Precalculus Review & Limits

Name: _____ Score: _____ / 100

Show all work. No calculators. No credit for using l'Hôpital's Rule.

1. **(6 pts)** Solve the inequality $|2x-3| > x+6$. Express the solution in interval notation.

2. **(8 pts)** Consider the points $P_1(-3,5)$ and $P_2(6,-1)$.
   (a) Find the equation of the line through P_1 and P_2.
   (b) Find the equation of the perpendicular bisector of the segment $\overline{P_1P_2}$.
   (c) Find the distance between P_1 and P_2 in simplest radical form.

3. **(8 pts)** Consider the function $f(x) = 3x^2 - x - 2$.
   (a) Evaluate $f(-2)$.
   (b) Evaluate and simplify $f(2+h)$.
   (c) Find all real x for which $f(x) = 0$.

4. **(8 pts)** Consider $f(x) = \dfrac{2x+1}{x-3}$ and $g(x) = \dfrac{3x+1}{x-2}$. Show that f and g are inverses of each other by verifying both $f(g(x)) = x$ and $g(f(x)) = x$. State the domain of f.

5. **(10 pts)** Let $g(x) = 3 - \sqrt{5 - x}$.

   (a) Determine the domain **D** and range **R** of g.

   (b) Sketch and label the graph, including the endpoint and both intercepts.

   (c) Find $g^{-1}(x)$ and state its domain and range.

6. **(8 pts)**

   (a) If $\csc(\alpha) = -\dfrac{13}{5}$ and $\cos(\alpha) > 0$, determine $\sin(\alpha)$, $\cos(\alpha)$, $\tan(\alpha)$, and $\sin(2\alpha)$.

   (b) Find all solutions of $2\sin^2 x + \sin x - 1 = 0$ in the interval $[0, 2\pi)$.

7. **(24 pts)** Evaluate the following limits using algebraic techniques (no credit will be awarded for using l'Hôpital's Rule):

   (a) $\displaystyle\lim_{x \to -1} \frac{2x^2 - x + 3}{x^3 + 5}$

   (b) $\displaystyle\lim_{x \to 3} \frac{x^3 - 27}{x^2 - 9}$

   (c) $\displaystyle\lim_{x \to 2} \frac{\sqrt{x + 7} - \sqrt{3x + 3}}{x - 2}$

   (d) $\displaystyle\lim_{\theta \to 0} \frac{1 - \cos(2\theta)}{\theta \sin(3\theta)}$

   (e) $\displaystyle\lim_{h \to 0} \frac{f(x + h) - f(x)}{h}$ for $f(x) = 2x^2 - 3x + 1$

   (f) $\displaystyle\lim_{h \to 0} \frac{f(x + h) - f(x)}{h}$ for $f(x) = \sqrt{2x + 1}$

8. **(10 pts)**

   (a) State the three conditions that must be satisfied for a function f to be continuous at $x = a$.

   (b) Find the value of the constant c that makes f continuous at $x = 2$, where

$$f(x) = \begin{cases} cx + 1, & x \leq 2 \\ x^2 - c, & x > 2. \end{cases}$$

9. **(8 pts)** Classify the discontinuity (removable, jump, or infinite) of each function at the indicated point, and justify using limits. For the removable one, state how the function should be redefined to remove the discontinuity.

   (a) $f(x) = \dfrac{x^2 - x - 6}{x - 3}$ at $x = 3$

   (b) $g(x) = \dfrac{|x|}{x}$ at $x = 0$

   (c) $h(x) = \dfrac{1}{(x-1)^2}$ at $x = 1$

10. **(10 pts)** Solve each equation.

    (a) $\log_2(x - 1) + \log_2(x + 1) = 3$

    (b) $2^{2x} - 5 \cdot 2^x + 4 = 0$

    (c) $\log_x 27 = \dfrac{3}{2}$